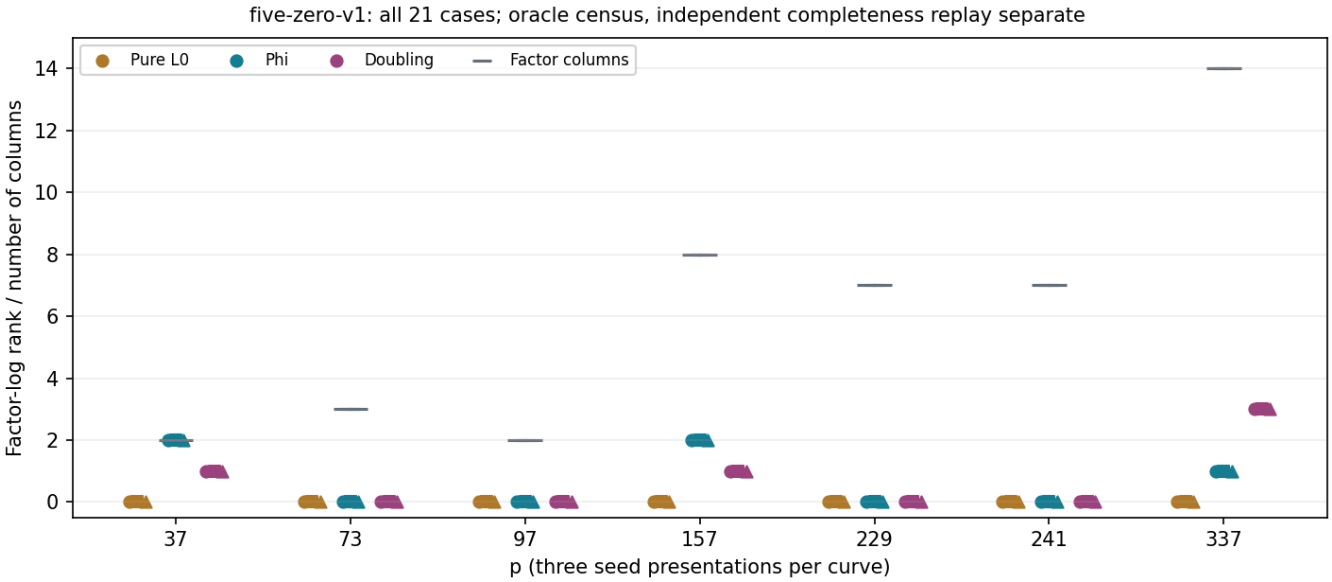
Five-factor zero census: equal aggregate counts, different rank

The native census completes all 21 cases and three arms, with zero missing cases. It reuses old point-verified matrix solutions as an oracle; it neither repeats BSGS nor supplies a polynomial relation-finding price. The required independent point-keyed completeness replay is a separate run.



Counts and factor-log rank

Pure five-ordinary-factor zero sets are empty on every case. Phi and doubling each yield 504 signed physical tuples and 15 distinct projective rows, summed over the 21 separate cases. Neither has an all-zero row in this five-factor census. Equal row counts do not imply equal rank.

p	Factor columns	Phi tuples / rank	Doubling tuples / rank
37	2	72 / 2	24 / 1
73	3	0 / 0	0 / 0
97	2	0 / 0	0 / 0
157	8	48 / 2	24 / 1
229	7	0 / 0	0 / 0
241	7	0 / 0	0 / 0
337	14	48 / 1	120 / 3

Every table row applies to each of its three retained seeds; all 21 records are preserved individually. At p37, phi's zero rows recover all two unanchored factor logs and verify every source/image point, while doubling reaches rank one. At p157 phi also has greater rank; at p337 doubling has greater rank. There is no uniform rank ordering over these curves. The seeds are not independent curve samples.

The pure and mixed joins check exactly ten and six partitions per physical tuple respectively, then deduplicate. Every retained five-point group sum is checked. The target coefficient is zero, and no target recovery is attempted. Full factor-log verification occurs only for the three p37 phi records; a rank-deficient result is not a failed point verification.

Cost and verification limits

The scalar-keyed enumeration records 35,456,448 pure triple queries and 107,972,352 for each mixed arm, plus pair-table sizes per case. These are oracle-work counters, not polynomial IC costs. All 79 native tests pass before the runner is copied. Source label is 11592c85b13d97dcffce3744424fa9db7a9d2665, with actual source and runner hashes. One worker, 8 GiB, 1,200 seconds per case and the busy reservation are retained.

Unchanged native summary: results/five-zero-v1-summary.json, SHA-256 14347fef57ee2bd1ed3ac43bc4b1efc3d0cd6365d4c9ea41d599a660ebf113d2. Its per-case hashes bind exact signed sets. The separate point replay contract (../../SUPPLEMENT_FIVE_POINT_REPLAY.md) uses public point keys, a unique canonical partition and another elimination algorithm. Until that replay passes, completeness remains independently unverified. Even a pass would still require a polynomial collector, charged setup, target descent, generic-image controls and complete matched prices before an end-to-end gain is claimed. No leaderboard or benchmark row is promoted.